

RESEARCH AI TASK 6

Machine Learning — Car Price Prediction

Linear Regression without scikit-learn

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1. Objective

The objective of this task is to predict the price of cars from the given Car Price Prediction dataset using linear regression. The implementation is done without using scikit-learn, as required by the task.

2. Dataset

The dataset contains 205 car records and 26 original columns. The target variable is price. The dataset contains numerical and categorical features.

3. Data Preparation

The car_ID column was not used as a predictive feature because it is an identifier. CarName was also not used because it contains individual car names and would create a high-cardinality feature. The remaining numerical features and useful categorical features were selected. Categorical features were converted into numerical form using one-hot encoding. The data was then divided into 80% training data and 20% testing data.

4. Feature Scaling

The numerical feature values were standardized using the mean and standard deviation calculated from the training data. The same training-data statistics were used to transform the test data. Feature scaling helps gradient descent work more efficiently when features have different numerical ranges.

5. Linear Regression Model

The model uses the linear regression form

$$\hat{y} = wx + b$$

where X contains the input features and W contains the parameters learned by the model. A bias column of ones was added to X so that the intercept is learned as part of W .

6. Gradient Descent

The parameters were initialized to zero and updated using gradient descent. For each iteration, predictions were calculated, the error was found, the gradient of the cost with respect to the parameters was calculated, and the parameters were updated using a learning rate of 0.01. The model was trained for 20,000 iterations.

7. Evaluation

The model was evaluated on the test set using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2

Quantities	Score
MSE	12,029,795.51
RMSE	3,468.40
R^2	0.829

The RMSE is in the same unit as the car price, while R^2 indicates the proportion of variation in the test-set prices explained by the model.

8. Result

The implemented linear regression model provides a reasonable baseline for predicting car prices. The complete process—from preprocessing to gradient descent and evaluation—can be understood and reproduced without relying on scikit-learn.

9. Conclusion

This task demonstrated the complete basic workflow of a supervised learning regression problem. The dataset was inspected, features were prepared, categorical values were encoded, the data was split into training and testing sets, features were scaled, and a linear regression model was trained using gradient descent. Finally, the model was evaluated using MSE, RMSE, and R^2 . The project provides a practical application of the linear regression and gradient descent concepts studied in the previous research task.

10. GitHub Repository

Link: https://github.com/mridulpoddar9/Mars_Rover_Manipal_MridulPoddar/tree/main
